Watershed Disturbance Mapping System

Technical Specification for Old Growth Forest Monitoring in the Pacific Northwest

Executive Summary

Automated detection system for fires, logging, and other disturbances in watersheds upstream of old growth forest stands. Achieves 85-92% operational accuracy using free satellite data with 2-week to 2-month detection lag in Pacific Northwest conditions.

Problem Statement

Manual analysis of 20+ years of satellite imagery across multiple watersheds requires months of human effort. Critical disturbances affecting downstream old growth habitats go undetected or are discovered too late for intervention.

System Capabilities

- **Detection Types**: Fire scars, clear-cut logging, selective logging, road construction, landslides, insect outbreaks
- **Minimum Detectable Size**: 0.04 hectares (400 m²) with optical, 0.02 hectares with radar
- **Temporal Coverage**: 1984-present (Landsat), 2015-present (Sentinel-2)
- **Update Frequency**: Monthly automated processing
- **Geographic Scale**: 100-10,000 km² watersheds

Data Architecture

Primary Satellite Platforms

Sentinel-2 (Recommended Baseline)

- Resolution: 10m (visible/NIR), 20m (red-edge/SWIR)
- Revisit: 5 days
- Spectral Bands: 13 including critical red-edge
- Cost: Free via Copernicus
- Cloud API: Google Earth Engine, Sentinel Hub

Harmonized Landsat Sentinel-2 (HLS)

- Combined temporal density: 2-3 day revisit
- Pre-processed surface reflectance
- Reduces detection lag from 3.3 to 0.22-2 months
- Seamless integration of both sensors

Sentinel-1 Radar (Cloud Penetration)

- Resolution: 10m
- All-weather capability
- Detects structural changes through clouds
- C-band SAR for canopy gap detection

Commercial Alternatives (When Justified)

Planet Labs PlanetScope

- Resolution: 3-5m daily
- Cost: $2,000-6,000/year per AOI
- Use Case: Critical watershed rapid response
- ROI: Early detection prevents downstream impacts

Machine Learning Architecture

Recommended: U-Net with ResNet50 Backbone

python

```python
import segmentation_models_pytorch as smp

model = smp.Unet(
    encoder_name="resnet50",
    encoder_weights="imagenet",
    in_channels=13,  # Sentinel-2 bands
    classes=5,  # Background, fire, logging, roads, landslides
    activation="softmax"
)
```

Performance Metrics

- Overall Accuracy: 90-95%
- Fire Detection: 94% (NBR threshold) to 99.91% (deep learning)
- Logging Detection: 86-92%
- User's Accuracy: 90.9% (excluding non-forest)
- Producer's Accuracy: 91.6%

Alternative Architectures

1. **DeepLabV3+**: Better multi-scale context (94-98% accuracy)
2. **Vision Transformers**: Temporal patterns (65-72% F1, experimental)
3. **Random Forest**: Baseline comparison (78-85% accuracy)

Change Detection Algorithms

Continuous Change Detection (CCDC)

- Harmonic regression modeling of time series
- F1-Score: 78.14%
- Temporal Accuracy: 51.8%

- Best for: High cloud cover scenarios

LandTrendr

- Temporal segmentation of pixel trajectories
- Overall Accuracy: 86.2%
- Best for: Distinguishing disturbance types
- Implementation: Google Earth Engine native

Multi-Sensor Fusion (FNRT)

- Combines Landsat + Sentinel-2 + Sentinel-1
- Early Detection: 69.8% within 30 days, 84.6% within 60 days
- Median Detection Lag: 100 days

Processing Pipeline

Monthly Workflow

bash

```bash
# Automated monthly processing
0. Schedule: Every 5 days (Sentinel-2 revisit)
1. Download new imagery →
2. Atmospheric correction (Sen2Cor) →
3. Cloud masking (s2cloudless, >30% rejection) →
4. Calculate indices (NDVI, NBR, TCG) →
5. Run inference (U-Net model) →
6. Change detection (compare to baseline) →
7. Validate detections (confidence >0.8) →
8. Update PostGIS database →
9. Generate alerts (email/SMS via Twilio)
```

Quality Control

- Cloud probability threshold: 30%
- Minimum clear observations: 3 per 90 days
- Confidence score filtering: >0.8 for alerts
- Spatial filtering: Remove detections <0.04 hectares
- Temporal persistence: Require 2 observations

Hardware Requirements

Minimum Development System

- GPU: NVIDIA RTX 3060 (12GB VRAM) - $400
- RAM: 16GB DDR4
- Storage: 500GB NVMe SSD
- Processing Time: 5-10 min per 100 km²

Production System

- GPU: RTX 3090/4090 (24GB VRAM) - $1,500-2,000
- RAM: 32-64GB
- Storage: 2TB NVMe + 10TB RAID
- Capabilities: 10-20 watersheds monthly

Cloud Alternative

- Google Earth Engine: Preprocessing
- AWS SageMaker: Training ($3/hour V100)
- Estimated Cost: $500-2,000/month operational

Implementation Tools

Core Python Stack

python

```python
# Installation
pip install torchgeo        # Geospatial deep learning
pip install rasterio gdal   # Raster I/O
pip install sentinelhub-py  # Sentinel data access
pip install segmentation-models-pytorch
pip install geopandas       # Vector operations
pip install earthengine-api # Google Earth Engine
```

Key Libraries

- **TorchGeo**: Pre-trained models, geospatial sampling
- **Rasterio**: Efficient GeoTIFF processing
- **WhiteboxTools**: Watershed delineation
- **Streamlit/Dash**: Dashboard deployment

Expected Accuracy & Limitations

Performance Benchmarks

- Clear conditions: 92-96% accuracy
- Cloudy conditions: 85-90% (with radar fusion)
- Small disturbances (<0.1 ha): 70-75% detection
- Large disturbances (>1 ha): 95%+ detection

Critical Failure Modes

1. **Cloud Cover**: 40-60% data loss in winter
    - Mitigation: Sentinel-1 radar fusion
2. **Snow Cover**: False change signals
    - Mitigation: Seasonal filtering
3. **Phenological Changes**: Deciduous false positives
    - Mitigation: Multi-year baseline
4. **Mixed Pixels**: 12.2% missed small gaps
    - Mitigation: Sub-pixel unmixing

Edge Cases

- Windthrow vs. logging: Requires temporal analysis
- Natural mortality vs. thinning: Spatial pattern analysis
- Shadows in mountainous terrain: Topographic correction
- Recent burns vs. harvest: Spectral trajectory analysis

Validation Strategy

Ground Truth Collection

- Initial: 100-150 labeled training patches
- Active Learning: 30-50 uncertain sites per iteration
- Target: 250-300 total labeled sites
- Field Validation: GPS + drone imagery

Metrics

- Overall Accuracy: Target >85%
- Kappa Coefficient: Target >0.80
- F1-Score per class: Target >0.75
- Spatial Cross-Validation: Block-based
- Temporal Holdout: Last 20% for testing

Timeline & Budget

Development Phase (6-9 months)

1. Data Pipeline Setup: Month 1-2
2. Model Development: Month 2-4
3. Field Validation: Month 3-6
4. Refinement: Month 5-7

5. Dashboard Deployment: Month 7-9

Costs

- Hardware: $2,000-5,000 (one-time)
- Cloud Services: $200-500/month
- Field Work: $5,000-10,000
- Total First Year: $15,000-25,000

Maintenance

- Model Retraining: Quarterly
- Validation Updates: Semi-annual
- Algorithm Updates: Annual
- Ongoing Cost: $3,000-6,000/year

Risk Assessment

Technical Risks

- Cloud cover preventing detection: MEDIUM
- False positive rate >15%: LOW
- Processing bottlenecks: LOW
- Model degradation over time: MEDIUM

Mitigation Strategies

- Multi-sensor fusion for cloud gaps
- Conservative confidence thresholds
- Distributed processing architecture
- Continuous learning framework

Success Criteria

- [ ] Detect 90% of disturbances >0.5 hectares
- [ ] Process monthly updates within 48 hours
- [ ] Maintain <10% false positive rate
- [ ] Generate actionable alerts within 2 weeks of event
- [ ] Reduce manual analysis time by 95%

Next Steps

1. Acquire Sentinel-2 data for pilot watershed
2. Implement baseline U-Net model
3. Establish field validation protocol
4. Deploy prototype dashboard
5. Iterate based on user feedback